ZOUBIR HIBA

Artificial Intelligence & Data Science Student

My Portfolio (Visit my portfolio)

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Address: Algeria | Chlef

Overview

Final-year Computer Science Engineering student at ESI Sidi Bel Abbès, specializing in Artificial Intelligence and Data Science. Hands-on experience in machine learning, deep learning, NLP, and data science through academic and personal projects, with additional background in software and backend development. Comfortable taking projects from problem formulation and experimentation through to implementation. Seeking to apply AI to real-world problems and pursue further research or industry experience in the field.

Skills

- **Machine Learning Frameworks** : PyTorch | TensorFlow | Scikit-learn | Hugging Face Transformers | OpenCV | MLflow | Airflow | Label Studio .
- **Models & Techniques :** CNN | U-Net | ResNet | Encoder–Decoder Architectures | Transformers | BERT | Diffusion Models | DDIM | OCR | Stable Diffusion | ControlNet | Pix2Pix | AnimateDiff | IP-Adapter .
- **Programming languages** :Python | Java | R | JavaScript | C .
- **Software Development & Data Engineering** : Pandas | NumPy | Node.js | Express.js | PostgreSQL | MongoDB | Hadoop | Apache Spark | Docker | Git | Linux
- **Languages** : French (B2) | English (Advanced) | Arabic (native).
- **Soft Skills :** Problem Solving | Analytical Thinking | Teamwork | Communication | Adaptability | Presentation Skills .

Education

Higher School of Computer Science 8 May 1945 - ESI SBA

Sidi Bel Abbes, Algeria **10/2022 - Present**

- **2022 – 2024 :** Preparatory classes (Algorithms, Algebra, Analysis, Probability, Computer Architecture, Office & Web...).
- **2024 – 2025 :** Engineering Core (Software Engineering, Operations Research, Language Theory, Databases, Networks...).
- **2024 – Present :** Specialization in AI & Data Science (ML, DL, Big Data, NLP, Software Engineering for Data Science...).

High School Chenoufi Mohamed

Sobha,Chlef, Algeria **09/2019 - 07/2022**

- Baccalaureate Diploma in Experimental Sciences Track — Ranked 1st in the school and commune with an average of **17.26/20 .**

Projects

Sign Language Production & Recognition System — AI & Computer Vision Project

GitHub

- Developed a bidirectional system for converting text/voice to sign-language video and sign language to text.
- Led the pose-to-video generation phase, mapping a custom 151-joint skeleton representation to MediaPipe Holistic and evaluating GAN- and diffusion-based approaches, including Pix2Pix, ControlNet + AnimateDiff, and ControlNet + IP-Adapter.
- Selected and optimized a **ControlNet + Stable Diffusion 1.5** pipeline using DDIM sampling, xFormers, attention slicing, and VAE tiling for execution on a single Tesla T4 GPU.

Predictive Maintenance — Association Rule Mining

GitHub

- Analyzed industrial sensor telemetry and error logs to identify patterns associated with equipment failures.
- Extracted **287 strong association rules** from ~876K records across 100 machines and benchmarked FP-Growth against Apriori, achieving a **3.5× speedup**.
- Developed an interactive Streamlit application for failure-risk assessment and pattern exploration.

Smart Notes Generator: Audio to Impact

GitHub

- Developed an audio-to-notes pipeline for transforming lecture recordings into structured summaries.
- Integrated **Whisper ASR** for transcription and fine-tuned/evaluated a **BART** summarization model on the SAMSum dataset.
- Built an interactive Streamlit interface for the complete transcription-to-summary workflow.

ESI PFE Management Platform — Full-Stack Web Application

GitHub

- Developed a web platform for managing final-year projects across Student, Teacher, and Admin workflows.
- Contributed as **Backend Developer**, implementing REST APIs, authentication, data validation, and database operations.

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NEEDLE — Tailor & Client Marketplace App

GitHub

- Developed a mobile marketplace connecting clients and tailors for customized clothing orders.
- Contributed as a **Flutter developer**, implementing authentication and key user-interface flows.

Equipment Display Annotation Pipeline — Computer Vision Project

- Contributed to an image annotation pipeline using **Label Studio** to create structured datasets for downstream object detection and OCR tasks.

Experience

IT Intern **Sonelgaz** *Chlef, Algeria* **09/2025 - 10/2025**

- Supported IT operations and maintenance at DESI, focusing on systems and database integration.